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Machine Learning Techniques

Unit-1 + PYQs

Topics

1. Machine learning ? Concept ? Application ? + 2 years.
2. Supervised, Unsupervised and Reinforcement learning. + 2 years.
3. Regression, classification and Clustering + 2 years.
4. Decision Tree learning.
5. Designing a learning System. + 2021-22
6. Genetic Algorithm
7. Support Vector Machine
8. Issues in Machine learning and data Science.

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Road to 5K

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AKTU- 2022-23, 2020-21 [10-marks; 2 marks]

Machine Learning

Key points:

- Subset of AI
- Use algorithms and statistical models.
- Improve performance on a specific task over time.
- Without being explicitly programmed
- The primary goal of is to teach computer
- Recognize patterns in data.
- Make predictions or decisions based on patterns.

Definition:

Machine learning is a field of artifial Intelligence that involves training computer systems to learn from data and improve their performance on a specific task without being explicitly programmed. The goal of ML is to enable computers to recognize patterns in data and make predictions or decisions based on these patterns.

learning refers to the process of training an algorithm to recognize patterns and make predictions based on data.

e.g → M.L. used for future prediction

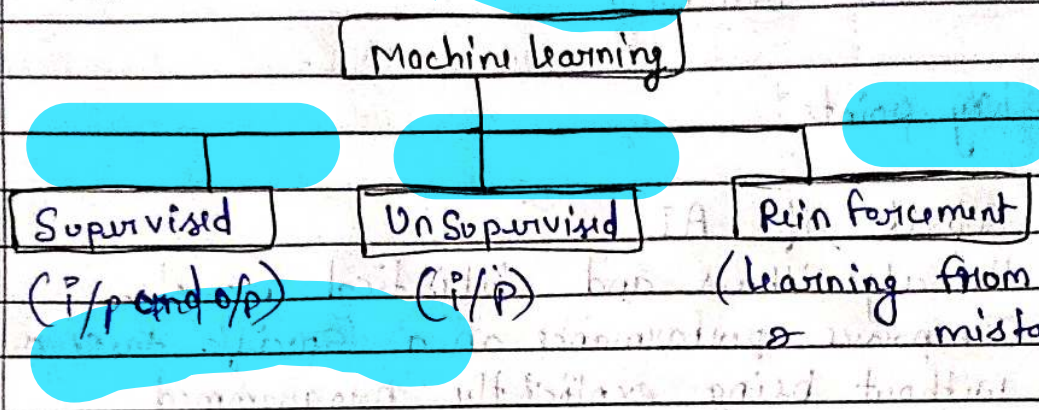
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AKtu - 2022-23

Types



① Supervised learning

→ It is the types of machine learning in which machines are trained using well "labelled" training data, and on basis of data, machine predict the output.

→ The labelled data means some input data is already tagged with the correct ouput.

→ In real world, Supervised learning used for
* Risk Assessment,
* Image Classification,
* Fraud Detection.

How Supervised learning works?

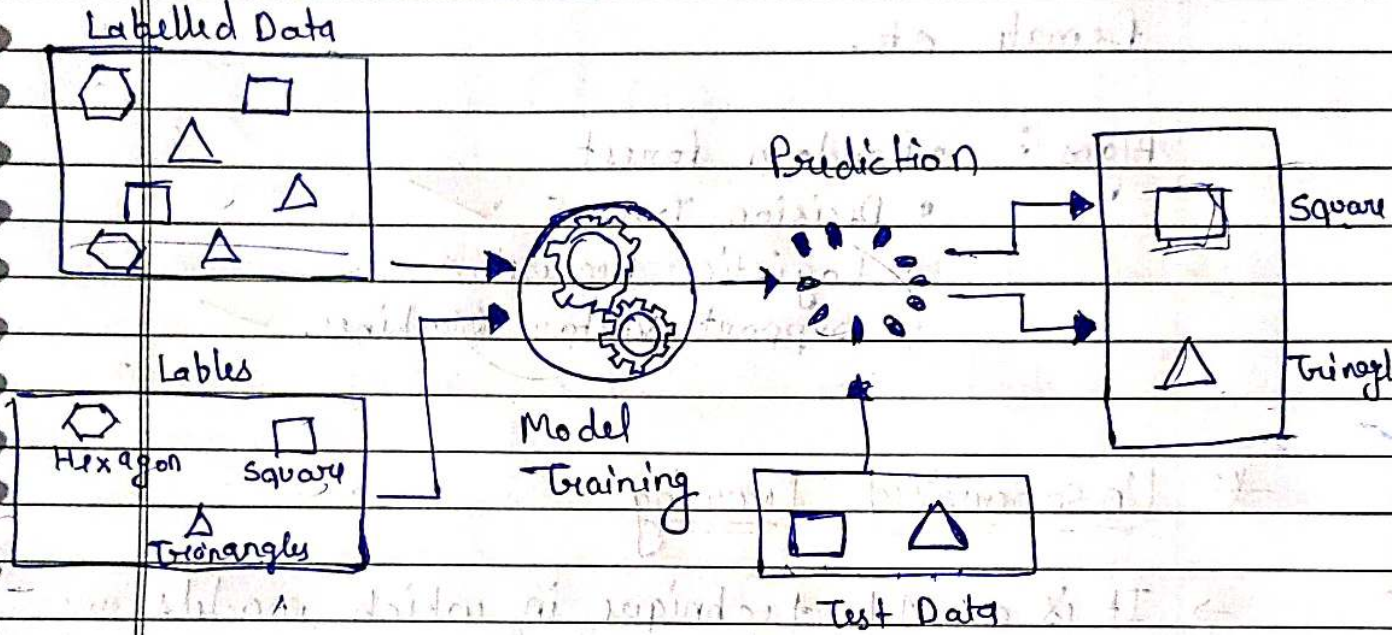
→ Suppose we have a dataset of diff. types of shapes which include square, rectangle, triangle and polygon. Now first step is that we need to train the model for each shape

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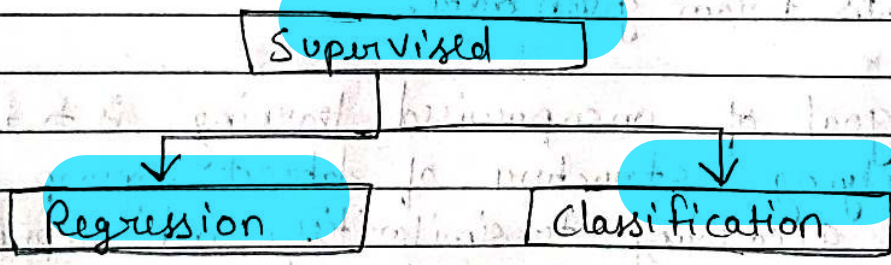
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6 If the given shape has four sides, and all the sides are equal ~~are~~ labelled as a Square
- If three sides labelled as triangle
- If six sides as hexagon.



Types of Supervised Learning. AKTU-2022-23



① Regression = It is used find relationship b/w the input variable and output variable. It is used for prediction of continuous variables, such as weather forecasting, Market Trends etc.

Algos :- Linear Regression, Regression Trees, Non-linear Regression, Bayesian Linear Regression.

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② Classification

Classification algorithms are used when the output variable is categorical, which means there are two classes such as Yes-No, male-female etc.

Algos :
- Random forest
- Decision Trees
- Logistic Regression
- Support Vector Machines.

* Unsupervised Learning

→ It is a ML technique in which models are not supervised using training dataset. Instead, models itself find the hidden patterns and insights from given data.

→ The goal of unsupervised learning is to find the underlying structure of dataset, group that data according to similarities, and represent that dataset in a compressed format.

→ Clustering : It is a method of grouping the objects into clusters such that objects with most similarities remains into a group and has less or no similarities with the objects of another group.

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Unsupervised Learning Algorithms :

- K-means clustering
- KNN
- Hierarchal clustering

- Neural Networks
- Anomaly detection.

Reinforcement Learning

→ A model is trained to make decisions based on trial and error.

→ It takes actions and receive feedback on those actions.

→ It receives rewards for making correct decisions and penalties for making incorrect decisions.

→ It helps to learn how to make better decision in the future.

→ Commonly used in the game playing, robotics, and Control System.

→ It is like children learn new behaviours and skills.

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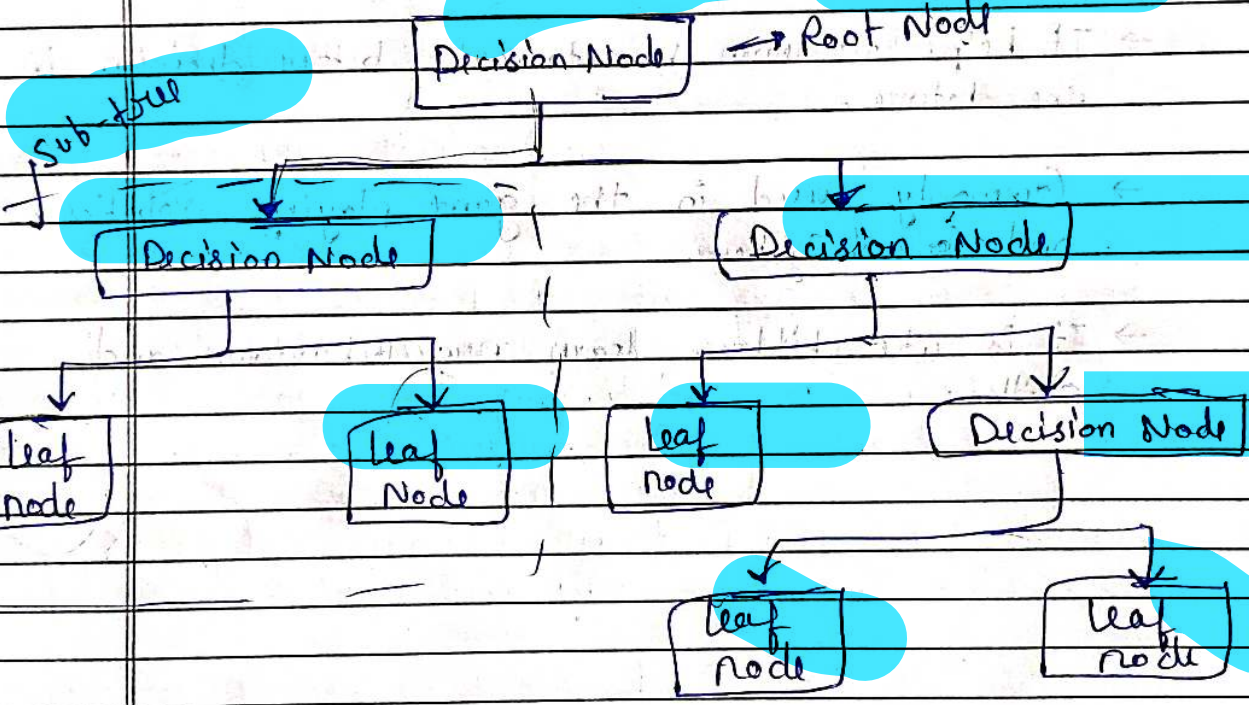
Decision Tree Learning

→ Decision Tree is a Supervised learning Technique that can be used for classification and Regression problems.

→ but mostly it is prefered for solving classification problems.

→ It is a tree-Structured classifier, where

① internal node → Features of a dataset
② branches → decision rules
③ leaf nodes → outcomes.

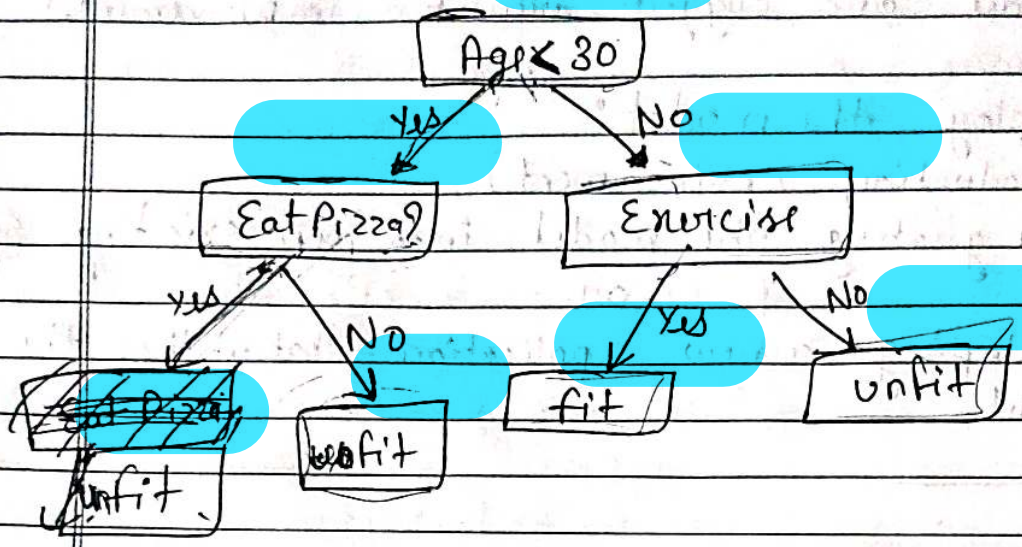


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e.g → Is person is fit or not.



* Designing a learning system → 1 year.

Steps :

① Define the problem :
(input, output and evaluation metrics).

② Collect and pre process data :
→ (data to train and test the system).
→ (involve cleaning, transforming and normalizing the data).

③ Choose an algorithm :
(Supervised or Unsupervised or Reinforcement).

④ Train the model :
→ (on the training data)
→ Recognize pattern.

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(5) Test the model:
(Identify any issues or limitations with the model and suggest areas for improvement)

(6) Deploy the model:
→ (production environment)
→ Integrating the model into an existing system or developing a new application that utilizes the model.

* Bayes' Theorem.

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

* Genetic Algorithm.

→ It is a type of optimization algorithm
→ It is inspired by the process of natural selection in biological evolution.
→ It starts by generating a bunch of random solutions and then tests them to see how well they perform.
→ The best solutions are then combined to create new solutions, just like new baby inherits traits from parents
→ The process is repeated until the best solution found
→ ~~Commonly used~~ → used for generate high-quality solutions for optimization problems and search problems.

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* Support Vector Machine

→ SVM is a type of supervised machine learning algorithm that can be used for classification or regression tasks.

→ SVM tries to find the best possible boundry between the different classes in the data by finding a hyper plane that maxmizes the margin b/w two two classes.

→ It is often used in applications such as image recognition, text classification etc.

* Issues in Machine Learning and data Science

① Data Quality:
→ essential to have good quality data to produce quality M.L algo and model.
→ Accuracy of M.L is depend on the quality of the data.

② Transparency:
→ It is difficult to make definitive statements on how well a model is going to generalize in new environments

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(3) Privacy and Security :

→ M.L Algos often require access of large amount of sensitive data, such as personal info, medical records or financial data

(4) Computational Resources : M.L algos can be computationally intensive and may require high amount of processing power or specialized hardware to run efficiently.

Unit-1 Completed

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